

Date: 01.01.2018 , Monday Time: 10.00 a.m. to 01.00 p.m Maximum Marks: 70

1. Q.1 and Q.4 are compulsory.
2. Figures to right indicate marks.
3. Section wise separate answer book to be used.
4. Draw neat sketches wherever necessary.

Q.1 **A)** Write a note on elution principle for column and planar chromatography. **05**
 B) Classify stationary phase used in Gas chromatography (GC) and explain the **05**
 factors governing the selection of stationary phase in GC.

- OR**

- Q.3** **Answer the following (Any three)** **15**
- (A) Explain validation of an analytical method.
- (B) Write a note on detectors used in HPLC
- (C) Differentiate between DSC and DTA.
- (D) Differentiate between GC and HPLC.

Q.5 (A) Discuss sample preparation methods used in IR. **05**
(B) Explain Bragg's law. Write applications of X-ray diffraction technique. **05**

OR

- Q.6** **Answer the following (Any three)** **15**

(A) Write note on FT-IR Technique.

(B) What is electrophoresis? Classify electrophoretic methods and discuss its basic principle.

(C) What are related substances and impurities? Discuss different techniques to identify impurities.

(D) Discuss the method of preparation and characterization of reference standard compound.

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